

Execution-order comparison

Pipeline: Normal LLM | 2 runs | 3 identical question texts | Reference: R2

Reference selected automatically: its actual execution sequence matches the selected questions in questionbank.xlsx sheet/row order.

Run	Folder / model
R1	b example-model
R2	a example-model

Main observations

These are observed differences, not a causal estimate of the effect of shuffling. Output length, background activity and measurement variation may contribute.

Recorded model names match; this does not establish identical model versions or generation settings.

Generation settings, token counts and RAG index identity are not consistently recorded by these reports; equality of those conditions cannot be verified.

Lowest observed emissions: R1, R2, 3 g CO₂e. Spread between highest and lowest: 0 g (0% relative to the lowest).

R1 versus R2: different execution sequence. 3 valid successful question pairs; 0 lower, 0 higher, 3 unchanged emissions.

Largest absolute change: Q3, 0 g; execution position 3 to 1. Net change over these matched successful pairs: 0 g.

R1: energy_wh change summed over 3 matched successful pairs: 0.

R1: latency_s change summed over 3 matched successful pairs: 0.

Recorded metadata differences: Ordering. Unknown or unrecorded values do not establish equivalence.

Historical zero defaults cannot be distinguished from measured zeros. Missing measurements are excluded and partial sums labelled. Percentage changes with a zero reference are N/A.

Recorded shuffling schemes

These rules and seeds come from saved ordering metadata, with CSV metadata used when available. Missing rules are not inferred from the observed sequence. A shuffled scheme can sometimes produce the same sequence.

Run	Recorded scheme	Seed
R1	Freely shuffle all selected questions	42
R2 (reference)	Keep original order	42

Execution sequence comparison

Positions 1-3. Reference: R2. Read down a column to follow that run. Read across a row to compare what ran at the same position.

Up arrow / blue: Earlier by N. Down arrow / amber: Later by N. Equals / grey: Same position. N counts positions relative to the reference. Colours show movement only.

Position	R2 - Reference Keep original order Seed: 42	R1 Freely shuffle all selected questions Seed: 42
1	= Q1 Reference	↑ Q3 Earlier by 2
2	= Q2 Reference	= Q2 Same position
3	= Q3 Reference	↓ Q1 Later by 2

Run cost and coverage

Observed sums include failed requests. Coverage is valid readings / all questions. Partial sums cannot be treated as complete batch costs.

Run	Emissions (g)	Energy (Wh)	Request time (s)	Failures / unknown
R1	3 [3/3]	3 [3/3]	6 [3/3]	0
R2	3 [3/3]	3 [3/3]	6 [3/3]	0

Batch costs: R1-R2

Emissions (g CO2e)

Observed sums across all requests; partial coverage is labelled.

R1		3
	Valid 3/3	
R2		3
	Valid 3/3	

Energy (Wh)

Observed sums across all requests; partial coverage is labelled.

R1		3
	Valid 3/3	
R2		3
	Valid 3/3	

Metadata differences

Different recorded values are marked *. Not recorded means unavailable, not equal.

Setting	R1	R2
Folder *	b	a
Model(s)	example-model	example-model
Pipeline	Normal LLM	Normal LLM
Ordering *	global	original
Seed	42	42
Questions	3	3
Runtime (s)	Not recorded	Not recorded
Completion evidence	No summary; completion unverified	No summary; completion unverified
codecarbon_version	Not recorded	Not recorded
os	Not recorded	Not recorded
effective_scope	Not recorded	Not recorded
measure_power_secs	Not recorded	Not recorded
pue	Not recorded	Not recorded
country_iso_code	Not recorded	Not recorded
config_profile	Not recorded	Not recorded
configured_settings	Not recorded	Not recorded
hardware	Not recorded	Not recorded
backends	Not recorded	Not recorded
hardware_evidence	Not recorded	Not recorded

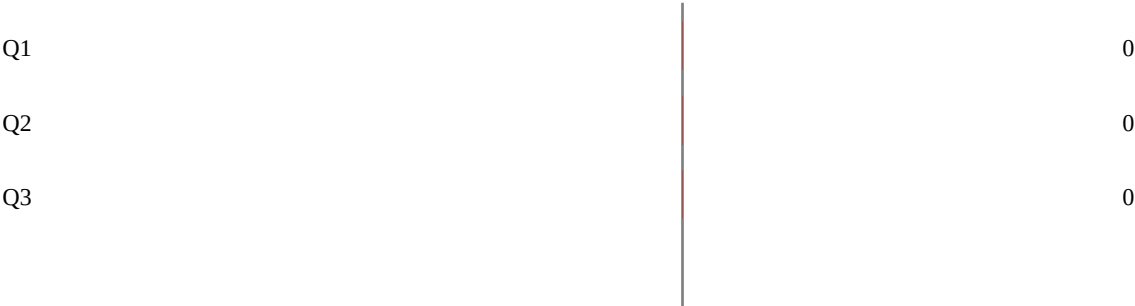
Question emissions change (%)

Relative to R2. Blue = lower, red = higher; grey = missing, failed, or zero reference. Q labels map to exact question text in the comparison table. Color saturates at +/-100%; cell numbers retain the actual change.

	R1	R2
Q1	0	0
Q2	0	0
Q3	0	0

Largest changes: R1 versus R2

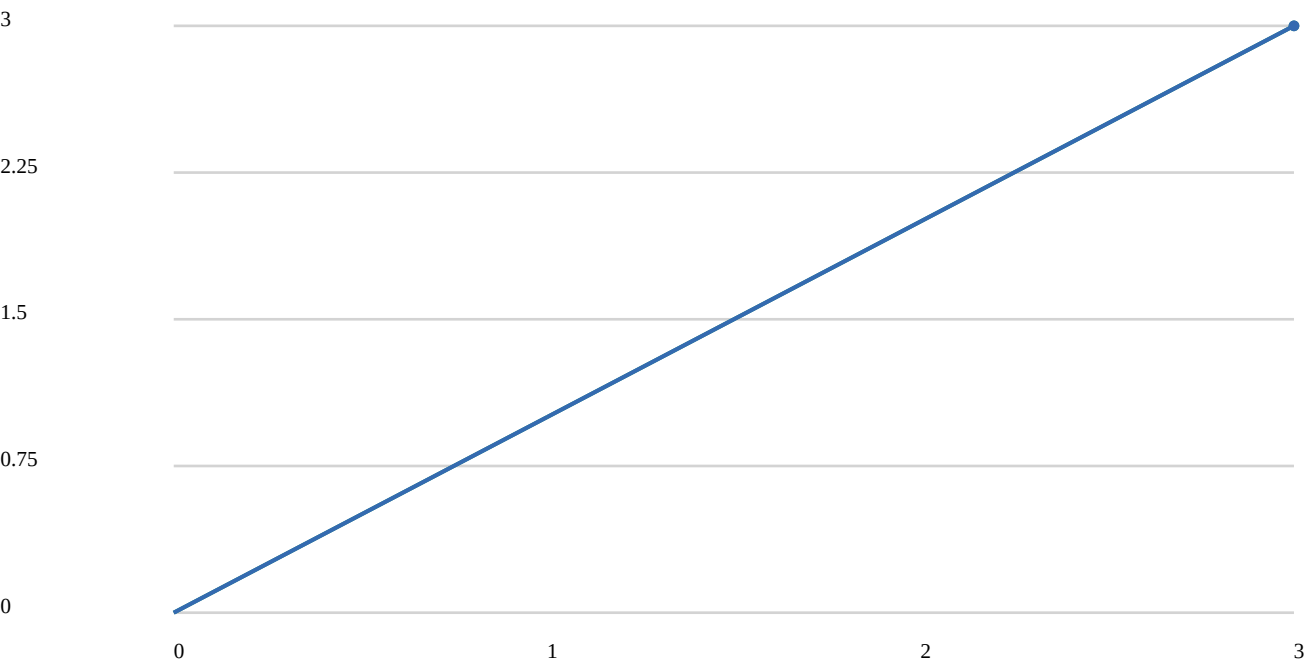
Absolute emissions changes for up to 12 matched successful questions. Left = decrease; right = increase. Labels show g CO2e.



Cumulative emissions through execution

X: number of questions executed. Y: cumulative g CO2e. Question identities at a given position can differ. Curves stop at the first missing measurement rather than implying zero cost. Failed-request measurements are included.

R1: 3/3 questions accumulated
R2: 3/3 questions accumulated



Aligned question comparison

Q identifiers follow the reference run. Text matching is exact. Mean emissions are calculated per question across all selected runs, using recorded values only; missing readings are excluded and the contributor count is shown. Recorded costs for failed attempts are included. Percentage changes are unavailable for a zero reference.

Q1: *What is a Python list?*

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	3	ok	1	1 (2/2)	1	3	0	0
R2	1	ok	1	1 (2/2)	1	1	0	0

Q2: *Explain compound interest.*

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	2	ok	1	1 (2/2)	1	2	0	0
R2	2	ok	1	1 (2/2)	1	2	0	0

Q3: *What is retrieval-augmented generation?*

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	1	ok	1	1 (2/2)	1	1	0	0
R2	3	ok	1	1 (2/2)	1	3	0	0

Question label key

Q labels follow the execution order in R2. The same exact question text has the same label in every run and throughout this PDF.

Label	Exact question text
Q1	What is a Python list?
Q2	Explain compound interest.
Q3	What is retrieval-augmented generation?